

VCF-CLIP: Visual Context-Driven Fine-Grained Prompt Learning for Zero-Shot Anomaly Detection

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Abstract—Benefiting from recent advances in vision-language models (VLMs), numerous CLIP-based zero-shot anomaly detection (ZSAD) methods have been proposed to address the cold-start problem. Despite their impressive performance, these methods still depend on manual prompt engineering, and their coarse-grained text prompts struggle to capture the diverse patterns of anomalies, resulting in suboptimal visual-text alignment. To overcome these limitations, we propose VCF-CLIP, a visual context-driven fine-grained prompt learning framework built upon CLIP. The novelties of VCF-CLIP lie in two main aspects. First, we propose the prompt prototype learning (PPL) strategy, which learns a pair of unified prompt prototypes representing general normal and anomalous states in a loss-guided manner, thereby eliminating the need for manual prompt design. Second, we propose a lightweight prompt refinement adapter that dynamically aggregates multiscale and multilevel visual features to iteratively refine the prompt prototypes, enabling the generation of instance-specific prompts enriched with fine-grained information. We conduct extensive experiments on 14 benchmarks across industrial and medical domains, and show that VCF-CLIP outperforms existing state-of-the-art ZSAD methods.

Index Terms—Adapter tuning, prompt learning, vision-language model (VLM), visual anomaly detection (AD), zero-shot learning.

I. INTRODUCTION

VISUAL anomaly detection (AD) is a critical task in computer vision, aiming to identify anomalous images and localize regions of anomaly [4]. It plays a vital role in domains such as industrial manufacturing [5], [6], [7], [8], and medical image analysis [9]. Although previous AD frameworks [10], [11], [12], [13] have shown promising performance, they suffer from the cold-start problem [2], as it is often infeasible to collect sufficient normal images for training on novel or rare classes. An effective solution is zero-shot AD (ZSAD), which aims to detect anomalies in unseen classes after training a unified model on auxiliary data from seen classes [1]. This

is a more practical yet challenging task, as it requires strong generalization capabilities when facing cross-domain datasets.

Recently, large-scale pretrained vision-language models (VLMs) [14], [15], [16], [17], such as CLIP [17], have shown strong zero-shot generalization capabilities across various visual downstream tasks [18], [19], inspiring the development of CLIP-based ZSAD methods [1], [2], [3], [20]. These methods typically formulate AD as a visual-text alignment task driven by textual prompts, making the design of effective prompts a key challenge [21], [22] in ZSAD.

In real-world scenarios, the wide variety of anomalies makes it challenging for a single textual prompt to cover all possible cases, leading to poor visual-text alignment. As shown in Fig. 1(a), a straightforward solution is prompt ensemble [1], [23], which manually designs prompt templates and constructs a large set of prompts based on domain-specific knowledge to improve coverage. However, manual prompt engineering is labor-intensive, and its effectiveness heavily depends on the expert knowledge. In addition, owing to the cognitive gap between humans and models, manually designed prompts are often suboptimal [21], [22]. To mitigate these issues, AdaCLIP [2] and AnomalyCLIP [3] propose prompt learning-based methods that utilize hybrid prompts consisting of both learnable and manually designed components. As shown in Fig. 1(b) and (c), AdaCLIP introduces learnable static prompts and dynamic prompts based on the visual class tokens to improve coverage of and adaptability to diverse anomalies, whereas AnomalyCLIP replaces the context words in textual prompts with learnable vectors to reduce the reliance on manual prompt engineering.

However, existing prompt learning-based methods suffer from two fundamental limitations: 1) they still rely on manual prompt engineering, and 2) the learned prompts lack fine-grained semantic information, which limits their ability to comprehensively capture the diverse visual patterns. To this end, we propose VCF-CLIP, a visual context-driven fine-grained prompt learning framework built on CLIP for ZSAD, as shown in Fig. 1(d). To eliminate the reliance on manual prompt engineering, we propose the prompt prototype learning (PPL) strategy, which learns a pair of unified prompt prototypes representing the general normal and anomalous states in a loss-guided manner. We employ an alignment loss for aligning prompt prototypes with visual representations and a regularization loss to introduce normal and anomalous concepts. In addition, we introduce the prompt refinement adapter (PR-Adapter), which iteratively refines the embeddings of prompt prototypes by dynamically aggregating multiscale and

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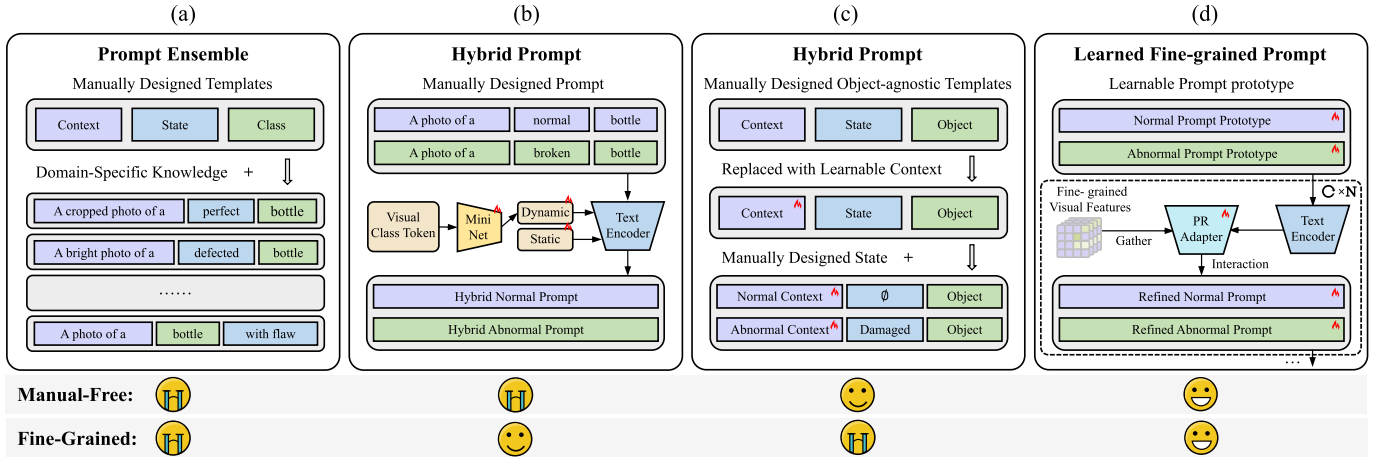


Fig. 1. Comparison of different prompt construction paradigms. (a) Prompt ensemble-based WinCLIP [1], (b) hybrid prompt-based AdaCLIP [2], (c) hybrid prompt-based AnomalyCLIP [3], and (d) our VCF-CLIP. Unlike prior methods, VCF-CLIP eliminates manual prompt engineering and the use of coarse-grained textual prompts. In particular, it first learns a pair of unified prompt prototypes via the PPL strategy, which are then iteratively refined by the PR-adaptor to generate instance-specific prompts enriched with the fine-grained information.

multilevel visual features through cross-attention, thereby generating instance-specific prompts enriched with fine-grained information.

Our main contributions are summarized as follows.

- 1) We propose VCF-CLIP, a visual context-driven prompt learning framework that generates instance-specific prompts with rich fine-grained information for ZSAD.
- 2) We propose the PPL strategy, which eliminates manual prompt engineering by learning a pair of unified prompt prototypes in a loss-guided manner.
- 3) We propose the PR-Adapter, a lightweight module that iteratively refines the prompt prototypes by aggregating multiscale and multilevel visual features.
- 4) Extensive experiments on 14 benchmarks spanning industrial and medical domains demonstrate that the VCF-CLIP outperforms existing state-of-the-art ZSAD methods.

II. RELATED WORK

A. Zero-Shot AD

Benefiting from the rapid advancements of foundation models, such as CLIP [17] and SAM [24], ZSAD has made significant progress. Existing ZSAD methods can be broadly categorized into two groups: training-free methods and training-required methods.

The training-free methods typically build directly on foundation models without any fine-tuning. For example, WinCLIP [1] carefully designs textual prompts to identify anomalies and introduces multiscale patch windows to improve anomaly localization (AL). SAA [23] integrates language-guided detection using Grounding DINO [25], followed by fine-grained segmentation through SAM. However, despite avoiding training, these methods often involve complex manual prompt engineering that relies on expert knowledge. Moreover, multi-stage processing further increases the computational overhead and memory usage during inference.

Training-required methods address ZSAD by leveraging annotated samples from seen classes as auxiliary data to adapt CLIP-based models. The APRIL-GAN [20] and CLIP-AD [26]

employ trainable linear layers to project patch-level features into a shared multimodal space, effectively eliminating the need for window-based operations and multistage processing. AnomalyCLIP [3] and AdaCLIP [2] further build on this foundation by introducing learnable class-agnostic prompts and dynamic prompts based on visual class tokens, respectively, to reduce reliance on manual prompt engineering and enhance visual-text alignment.

Despite these advances, all these methods still rely on manual prompt engineering and employ coarse-grained textual prompts. On the contrary, VCF-CLIP eliminates manual prompt engineering and integrates CLIP's inherent fine-grained visual information into textual prompts to generate instance-specific prompts.

B. PEFT for VLMs

With the rapid advancement of large language models, efficient transfer learning has attracted increasing attention from the research community [27], [28], [29]. Parameter-efficient fine-tuning (PEFT) [30], [31], [32], also referred to as Delta-tuning [33], aims to improve the fine-tuning efficiency by updating only a small subset of parameters. Although initially developed for natural language processing, PEFT has also been widely explored in VLMs.

In VLMs, PEFT methods mainly fall into two categories: prompt learning and adapter tuning. CoOp [22] first introduced prompt learning to VLMs by replacing the context words in prompts with learnable vectors, thereby improving performance on downstream tasks. Building on CoOp, CoCoOp [21] generates image-conditioned prompts to enhance generalization to unseen classes, while VPT [34] inserts learnable tokens into the visual encoder. By contrast, adapter tuning introduces lightweight trainable modules into pretrained models, enabling efficient adaptation with minimal modification to the original parameters. For example, LoRA [30] incorporates trainable low-rank matrices to enable efficient fine-tuning without updating the pretrained weights. The CLIP-Adapter [35] adopts a residual-style feature blending mechanism for few-shot transfer learning, while Tip-Adapter [36] further

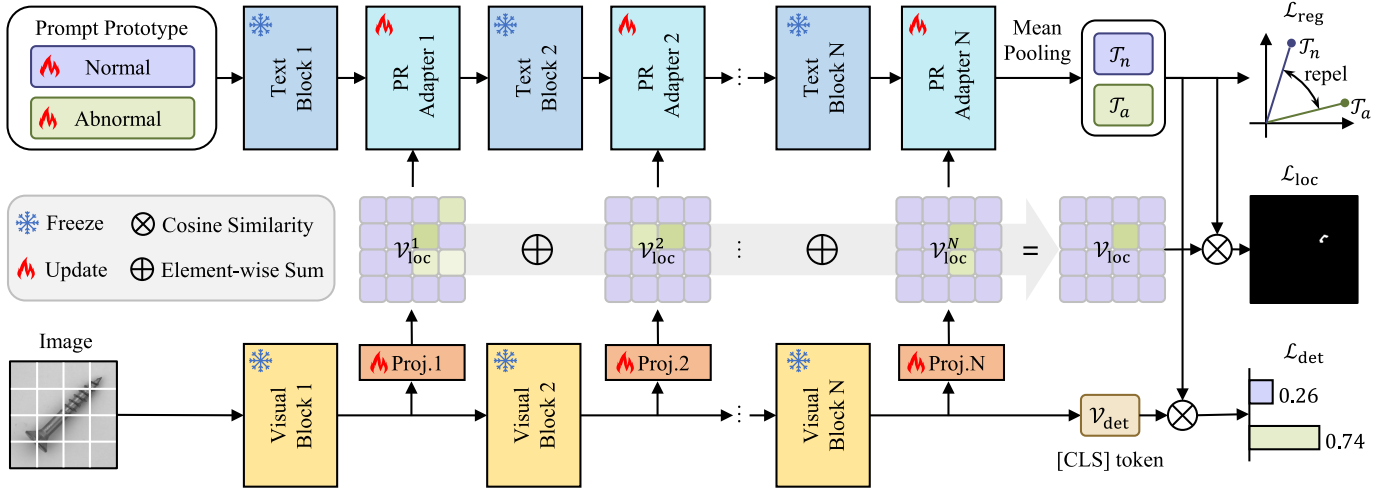


Fig. 2. Overview of the proposed VCF-CLIP framework, which integrates the PPL strategy and the PR-adapter. The PPL strategy learns a pair of unified prompt prototypes in a loss-guided manner, thereby eliminating reliance on manual prompt engineering, while the PR-Adapter is embedded within the text encoder to iteratively refine the embeddings of these prototypes by dynamically aggregating multiscale and multilevel visual features.

introduces a feature caching strategy to improve the few-shot performance of VLMs.

While previous methods typically employ prompt learning and adapter tuning as independent modules to enhance the performance of VLMs on the downstream tasks, such as viewpoint invariance [37], semantic segmentation [18], and visual AD [2], [3], [38], we propose a novel framework that integrates both approaches to generate fine-grained textual prompts for ZSAD.

III. METHOD

A. Problem Formulation

The objective of ZSAD is to develop a universal AD model capable of detecting anomalies across diverse domains without training on the target dataset. Thus, a reasonable assumption is that the object categories in the test set are unseen during training [3]. Let $\mathcal{D}_{\text{aux}} = \{(X_j, Y_j, y_j)\}_{j=1}^J$ be an auxiliary training dataset consisting of J samples, including both normal and anomalous images. Here, $X_j \in \mathbb{R}^{h \times w \times 3}$ denotes the j th image, $Y_j \in \mathbb{R}^{h \times w}$ is the corresponding pixel-level anomaly mask, and $y_j \in \{0, 1\}$ is the image-level label, with $y_j = 0$ for normal and $y_j = 1$ for anomaly. The test set $\mathcal{D}_{\text{test}}$ may encompass multiple domains with varied object categories and anomaly types. Notably, the categories in the auxiliary training and test datasets, $\mathcal{C}_{\text{aux}}$ and $\mathcal{C}_{\text{test}}$, are disjoint (i.e., $\mathcal{C}_{\text{aux}} \cap \mathcal{C}_{\text{test}} = \emptyset$), which is characteristic of the zero-shot setting. The goal is to predict both image-level and pixel-level anomaly labels for any test image.

B. Overview of VCF-CLIP

As illustrated in Fig. 2, VCF-CLIP is a dual-branch model built on CLIP, comprising a visual branch and a textual branch. Given an input image $X \in \mathbb{R}^{h \times w \times 3}$, the visual encoder extracts features and applies linear projections to align the embedding dimension with that of the textual branch. This process yields a set of multilevel patch embeddings $\{V_{\text{loc}}^i \in \mathbb{R}^{(h/p) \times (w/p) \times d} \mid i = 1, \dots, N\}$ and a global class token $V_{\text{det}} \in \mathbb{R}^d$, where N is

the number of encoder blocks, p is the patch size, and d is the embedding dimension. Prompt prototypes describing normal and anomalous states, P_n and P_a , are encoded by the text encoder and iteratively refined by the PR-adapter. The final textual embeddings, $T_n, T_a \in \mathbb{R}^d$, are then obtained by mean pooling over all token embeddings. The AD score $\hat{y} \in \mathbb{R}$ and the AL map $\hat{Y} \in \mathbb{R}^{h \times w}$ are then computed by measuring the similarity between visual and textual embeddings.

The key innovation of VCF-CLIP lies in two components: the PPL strategy, which completely eliminates the reliance on manual prompt engineering, and the PR-adapter, which dynamically aggregates visual features to produce instance-specific fine-grained prompts. A detailed exposition of these two components is provided below.

C. PPL Strategy

Previous textual prompt designs typically rely on manually crafted context words, state descriptors, and class labels [1], [2], [3], [20], [23], each of which requires domain-specific expertise. On the contrary, our PPL strategy learns a pair of unified prompt prototypes in a loss-guided manner, thereby eliminating the need for handcrafted components. In particular, both the normal and anomalous prompt prototypes in our approach consist exclusively of soft prompt tokens, defined as follows:

$$P_n = [p_n^1] [p_n^2], \dots, [p_n^L] \quad P_a = [p_a^1] [p_a^2], \dots, [p_a^L] \quad (1)$$

where each $p_n^l, p_a^l \in \mathbb{R}^d$ denotes the l th prompt token embedding for the normal and anomalous states, respectively, and L is the prompt length.

In previous methods, the concepts of normal and anomalous states are introduced into prompts via state descriptors such as “good” and “damaged.” As our prompts are entirely free from manual design, we introduce a regularization loss $\mathcal{L}_{\text{reg}}$ to incorporate the concepts of normal and anomalous into the prompts, which is defined as

$$\mathcal{L}_{\text{reg}} = \left\langle T_n, T_a \right\rangle. \quad (2)$$

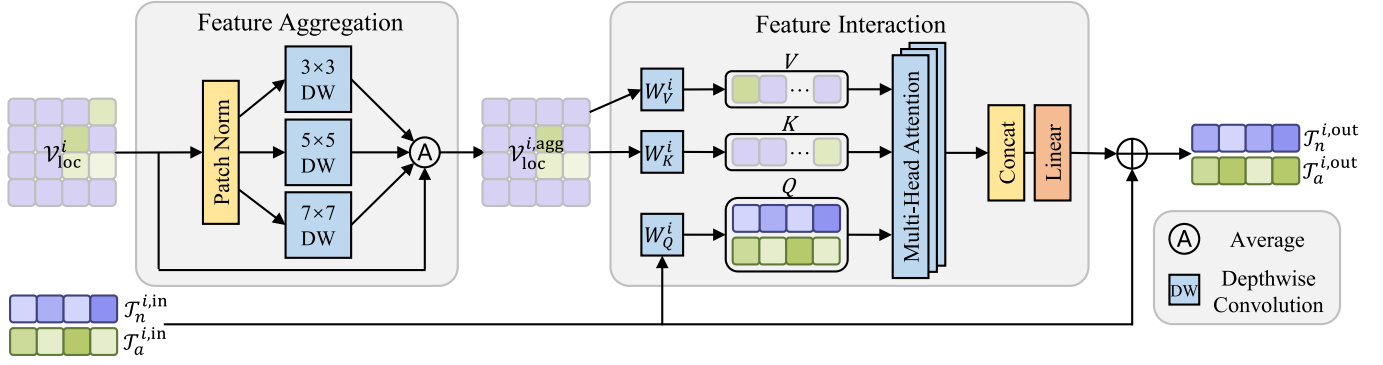


Fig. 3. Detailed pipeline of the PR-Adapter. It consists of a feature aggregation module that integrates multiscale visual features and a feature interaction module that refines text prompts by injecting visual information via cross-attention.

Here, $\langle \cdot, \cdot \rangle$ denote cosine similarity. The intuition behind the regularization loss is that normal and anomalous are mutually exclusive concepts; therefore, T_n and T_a should be distinctly separated in the latent space.

In addition, to ensure effective alignment between the textual representations and their corresponding visual representations, we introduce an alignment loss $\mathcal{L}_{\text{ali}}$, which consists of a detection loss $\mathcal{L}_{\text{det}}$ and a localization loss $\mathcal{L}_{\text{loc}}$

$$\mathcal{L}_{\text{ali}} = \underbrace{\text{BCE}(\hat{y}, y) + \text{Focal}(\hat{y}, y)}_{\mathcal{L}_{\text{det}}} + \underbrace{\text{Dice}(\hat{Y}, Y) + \text{Focal}(\hat{Y}, Y)}_{\mathcal{L}_{\text{loc}}}. \quad (3)$$

In particular, $\mathcal{L}_{\text{det}}$ combines binary cross-entropy (BCE) and focal loss [39] to supervise the AD predictions $\hat{y}$, while $\mathcal{L}_{\text{loc}}$ employs dice loss [40] and focal loss to guide the AL predictions $\hat{Y}$, where y and Y denote the ground-truth label and mask, respectively. With the joint supervision of $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{ali}}$, the unified prompt prototypes are learned in a fully automated manner, without any manual prompt engineering.

D. PR-Adapter

Although the PPL strategy eliminates the reliance on manual prompt engineering, the unified prompt prototypes it learns lack fine-grained information, thereby limiting their ability to capture the full diversity of anomalies and resulting in sub-optimal alignment with the visual representations. To address this limitation, we propose the PR-Adapter, which iteratively refines the prompt prototypes.

As illustrated in Fig. 3, the PR-adapter comprises a feature aggregation module and a feature interaction module. The feature aggregation module applies three depthwise-separable convolutions with distinct kernel sizes to aggregate multiscale visual features while remaining lightweight. Given the patch embeddings V_{loc}^i from the i th visual encoder block, it produces the aggregated features $V_{\text{loc}}^{i,\text{agg}}$

$$V_{\text{loc}}^{i,\text{agg}} = \frac{1}{4} \left[V_{\text{loc}}^i + \sum_{k \in \{3,5,7\}} \mathcal{D}_k(\mathcal{N}(V_{\text{loc}}^i)) \right] \quad (4)$$

where $\mathcal{N}(\cdot)$ denotes LayerNorm and $\mathcal{D}_k(\cdot)$ represents a depthwise-separable convolution with kernel size k .

Subsequently, $V_{\text{loc}}^{i,\text{agg}}$, together with the textual embeddings $T_n^{i,\text{in}}$ and $T_a^{i,\text{in}}$ from the corresponding i th text encoder block, is fed into the feature interaction module. In particular, $V_{\text{loc}}^{i,\text{agg}}$ is linearly projected by W_K^i and W_V^i to serve as the key and value, while $T_n^{i,\text{in}}$ and $T_a^{i,\text{in}}$ are projected by W_Q^i to obtain the queries for the cross-attention mechanism

$$q_*^i = W_Q^i T_*^{i,\text{in}}, \quad k^i = W_K^i V_{\text{loc}}^{i,\text{agg}}, \quad v^i = W_V^i V_{\text{loc}}^{i,\text{agg}} \quad (5)$$

where $* \in \{n, a\}$ denotes the normal and anomalous states. These queries, keys, and values are then split into M heads for multihead attention computation

$$H_{*,m}^i = \text{Softmax} \left(\frac{q_{*,m}^i (k_m^i)^\top}{\sqrt{d_{\text{head}}}} \right) v_m^i \quad (6)$$

$$O_*^i = \text{Concat}(H_{*,1}^i, \dots, H_{*,M}^i) W_O^i \quad (7)$$

where W_O^i is the output projection matrix, m indexes the attention heads, and d_{head} is the hidden dimension of each head. Finally, the refined prompt embedding is given by

$$T_*^{i,\text{out}} = T_*^{i,\text{in}} + O_*^i. \quad (8)$$

We insert a PR-adapter after each text encoder block to aggregate multilevel visual features, thereby iteratively refining the representations of the prompt prototypes.

E. Training and Inference

The model is trained on an auxiliary dataset using a loss function that combines the alignment loss $\mathcal{L}_{\text{ali}}$ and the regularization loss $\mathcal{L}_{\text{reg}}$, formulated as

$$\mathcal{L} = \mathcal{L}_{\text{ali}} + \lambda \mathcal{L}_{\text{reg}} \quad (9)$$

where λ is a hyperparameter controlling the weight of the regularization loss. The detailed definitions of $\mathcal{L}_{\text{ali}}$ and $\mathcal{L}_{\text{reg}}$ are provided in (3) and (2), respectively.

The AD score $\hat{y}$ is computed by measuring the similarity between the global visual embedding and the textual embeddings for anomalous and normal concepts

$$\hat{y} = \frac{\exp(\langle T_a, V_{\text{det}} \rangle)}{\exp(\langle T_a, V_{\text{det}} \rangle) + \exp(\langle T_n, V_{\text{det}} \rangle)}. \quad (10)$$

For AL, we first aggregate the patch embeddings from all N visual encoder blocks

$$V_{\text{loc}} = \frac{1}{N} \sum_{i=1}^N V_{\text{loc}}^i \quad (11)$$

and then compute the AL map as

$$\hat{Y} = \phi \left(\frac{\exp((T_a, V_{\text{loc}}))}{\exp((T_a, V_{\text{loc}})) + \exp((T_n, V_{\text{loc}}))} \right) \quad (12)$$

where $\phi(\cdot)$ denotes bilinear interpolation upsampling.

IV. EXPERIMENT

A. Experimental Setup

1) *Datasets*: To evaluate the proposed VCF-CLIP, we conduct experiments on 14 public real-world datasets spanning industrial and medical domains. For industrial AD, we adopt seven standard benchmarks: MVTec-AD [41], VisA [42], MPDD [43], BTAD [44], KSDD [45], DAGM [46], and DTD-Synthetic [47]. In the medical domain, seven datasets are considered, covering diverse diagnostic tasks. In particular, HeadCT [48], BrainMRI [49], and Br35H [50] are used for brain tumor classification; ISIC [51] for skin cancer detection; ClinicDB [52] and ColonDB [53] for colon polyp segmentation; and TN3K [54] for thyroid nodule detection. Following prior works [2], [20], [55], MVTec-AD is used as the default auxiliary training set, and we directly evaluate on the remaining industrial datasets as well as all medical datasets. For MVTec-AD itself, VisA serves as the auxiliary training set.

2) *Metrics*: Following prior works [1], [2], [3], [55], we use the area under the receiver operating characteristic curve (AUROC) as the evaluation metric. For the seven industrial datasets with both image- and pixel-level annotations, we report the AUROC at both levels to comprehensively assess performance in AD and AL. On the contrary, the medical datasets differ in label availability: ISIC, ColonDB, ClinicDB, and TN3K contain only anomalous test images, making image-level AUROC inapplicable, while HeadCT, BrainMRI, and Br35H lack pixel annotations, precluding pixel-level evaluation.

3) *Implementation Details*: For a fair comparison with existing methods [2], [3], [20], we adopt the pretrained CLIP (ViT-L/14@336px) released by OpenAI [56] as the default backbone. All input images are resized to 518×518 . Since the backbone's native input resolution is 336×336 , we resize its positional embeddings via bicubic interpolation to match the 518×518 inputs, following prior works [2], [3], [20]. In our design, the text and visual encoders of CLIP are evenly partitioned into four blocks, and the visual features from the sixth, 12th, 18th, and 24th layers are used for AL, consistent with prior work [2], [3], [20]. All CLIP parameters remain frozen except for the final projection layer of the visual encoder. The hyperparameter λ in the loss function is set to 0.1 by default. The number of attention heads M in the PR-Adapter, and the prompt length L are both set to 16. The model is trained for five epochs using AdamW [57] with a learning rate of 1×10^{-4} and a batch size of 16. All experiments are

conducted on a Linux machine equipped with an Intel¹ Xeon¹ CPU E5-2678 v3 @ 2.50 GHz and a single NVIDIA GeForce RTX 3090 GPU.

B. Main Experimental Results

1) *Comparison Methods*: We compare our method against several state-of-the-art ZSAD models. Among them, WinCLIP [1] and SAA [23] are training-free, while APRIL-GAN [20], AdaCLIP [2], AnomalyCLIP [3], and AA-CLIP [58] require training on the auxiliary datasets. Since AA-CLIP adopts VisA as its default auxiliary dataset, while AdaCLIP leverages additional auxiliary datasets, we retrain both methods using their official implementations under the same auxiliary-data setting as APRIL-GAN, AnomalyCLIP, and our method to ensure a fair comparison.

Formally, our method is related to AdaCLIP [2] and VCP-CLIP [59], as all three methods use visual context to refine prompts. However, VCF-CLIP differs from them in two main aspects. First, AdaCLIP and VCP-CLIP follow the hybrid-prompt paradigm and still require manually designed prompts for initialization. These prompts rely heavily on expert knowledge and may be suboptimal because of the cognitive gap between humans and models [3], [58]. On the contrary, VCF-CLIP removes the need for manual prompt engineering through the proposed PPL strategy. By directly optimizing the prompts for the AD task, PPL allows the model to automatically learn suitable prompt pairs without restricting them to natural-language expressions. Second, the use of visual context in existing methods is still limited. The AdaCLIP only uses the class token, which lacks fine-grained visual details, while VCP-CLIP uses multilevel visual features but only at a single scale. Moreover, both methods refine the prompt only once, which prevents sufficient integration of visual information into the textual prompts. The VCF-CLIP addresses these limitations with the proposed PR-Adapter, which progressively integrates multiscale and multilevel visual context. The aggregated visual context is then further fused through CLIP's native text encoder. In this way, VCF-CLIP improves visual-text alignment and better captures diverse anomaly patterns.

2) *Quantitative Results*: The quantitative results are presented in Tables I and II. VCF-CLIP demonstrates superior performance in both AL and AD tasks across diverse industrial and medical datasets.

For AL, VCF-CLIP achieves the highest overall average pixel-level AUROC of 92.4%, surpassing the strongest baseline, AnomalyCLIP, by 1.7%. In the industrial domain, it achieves the best pixel-level AUROC on every dataset and attains the highest average pixel-level AUROC of 96.6%. In the medical domain, VCF-CLIP also remains competitive, ranking first or second on all benchmarks and achieving the highest average AUROC of 84.9%. These results indicate that the learned fine-grained prompts in VCF-CLIP significantly enhance patch-level visual-text alignment within the CLIP framework.

For AD, VCF-CLIP also achieves the highest overall results, with an average image-level AUROC of 90.6%, outperforming AnomalyCLIP by 1.3% and AA-CLIP by 0.6%. Furthermore,

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TABLE I

PIXEL-LEVEL AUROC (%) OF ZSAD METHODS IN INDUSTRIAL AND MEDICAL DOMAINS. METHODS MARKED WITH [†] ARE TRAINING-FREE, WHILE THE OTHERS REQUIRE TRAINING. THE BEST PERFORMANCE IS IN BOLD, AND THE SECOND-BEST IS UNDERLINED

Domain	Dataset	SAA [†] [23]	WinCLIP [†] [1]	APRIL-GAN [20]	AdaCLIP [2]	AnomalyCLIP [3]	AA-CLIP [58]	VCF-CLIP
Industrial	MVTec-AD	75.5	85.1	87.6	90.0	91.1	91.9	92.5
	VisA	76.5	79.6	94.2	96.3	95.5	95.5	96.3
	MPDD	81.7	76.4	94.1	96.0	96.5	96.2	96.5
	BTAD	65.8	72.7	60.8	94.0	94.2	94.0	94.8
	KSDD	78.8	68.8	79.8	<u>95.8</u>	90.6	90.0	98.6
	DAGM	62.7	87.6	82.4	95.0	95.6	91.6	98.6
	DTD-Synthetic	76.7	83.9	95.3	<u>98.5</u>	97.9	96.4	99.2
Average		74.0	79.2	84.9	<u>95.1</u>	94.5	93.7	96.6
Medical	ISIC	83.8	83.3	89.4	86.1	89.7	93.9	92.9
	ColonDB	71.8	70.3	78.4	82.4	81.9	81.6	<u>82.2</u>
	ClinicDB	66.2	51.2	80.5	81.5	82.9	81.3	<u>81.9</u>
	TN3K	66.8	70.7	73.6	78.5	81.5	<u>81.7</u>	82.5
Average		72.2	68.9	80.5	82.1	84.0	<u>84.6</u>	84.9
Overall Average		73.3	75.4	83.3	90.4	<u>90.7</u>	90.4	92.4

TABLE II

IMAGE-LEVEL AUROC (%) OF ZSAD METHODS IN INDUSTRIAL AND MEDICAL DOMAINS. METHODS MARKED WITH [†] ARE TRAINING-FREE, WHILE THE OTHERS REQUIRE TRAINING. THE BEST PERFORMANCE IS IN BOLD, AND THE SECOND-BEST IS UNDERLINED

Domain	Dataset	SAA [†] [23]	WinCLIP [†] [1]	APRIL-GAN [20]	AdaCLIP [2]	AnomalyCLIP [3]	AA-CLIP [58]	VCF-CLIP
Industrial	MVTec-AD	63.5	91.8	86.1	90.3	<u>91.5</u>	90.5	91.1
	VisA	67.1	78.1	78.0	76.1	82.1	84.6	<u>82.9</u>
	MPDD	42.7	63.6	73.0	74.3	77.0	77.4	78.3
	BTAD	59.0	68.2	73.6	<u>93.3</u>	88.3	93.8	88.5
	KSDD	68.6	84.3	79.8	<u>95.1</u>	84.7	92.3	97.3
	DAGM	87.1	91.8	94.4	<u>94.4</u>	97.5	93.9	<u>96.5</u>
	DTD-Synthetic	<u>94.4</u>	93.2	86.4	95.2	93.5	93.3	95.2
Average		68.9	81.6	81.6	88.4	87.8	<u>89.4</u>	90.0
Medical	HeadCT	46.8	81.8	89.1	92.3	93.4	<u>92.5</u>	90.5
	BrainMRI	34.4	86.6	89.3	86.0	90.3	<u>91.8</u>	92.5
	Br35H	33.2	80.5	93.1	81.4	94.6	89.4	<u>93.4</u>
	Average	38.1	83.0	90.5	86.6	92.8	91.2	<u>92.1</u>
Overall Average		59.7	82.0	84.3	87.8	89.3	<u>90.0</u>	90.6

VCF-CLIP achieves the best AD results on four datasets, with average image-level AUROC scores of 90.0% and 92.1% in the industrial and medical domains, respectively. Moreover, compared with AnomalyCLIP, which also employs learnable context words, VCF-CLIP achieves larger gains on datasets with relatively small anomalous regions, such as VisA, KSDD, MPDD, and DTD-synthetic, while showing limited improvements on datasets with larger defects, such as MVTec-AD and BTAD. This may be because small anomalies require more precise patch-level visual-text alignment, for which the learned fine-grained prompts in VCF-CLIP are more effective. By contrast, for large and salient defects, a relatively coarse textual description is often sufficient to capture the anomaly.

Overall, these results highlight the strong effectiveness and generalization capability of VCF-CLIP in both the AL and AD tasks across diverse domains.

3) *Visualization*: To provide an intuitive demonstration of the AL capability of VCF-CLIP, we conduct a qualitative comparison with state-of-the-art methods, including AdaCLIP [2] and AnomalyCLIP [3], as shown in Fig. 4. It can be observed that both AdaCLIP and AnomalyCLIP successfully identify anomalies at the image level; however, our VCF-CLIP

produces fewer false negatives across both industrial and medical domains, accurately localizing the true anomalous regions with sharper and more consistent boundaries. This observation is consistent with the quantitative results, where VCF-CLIP shows a significant improvement in AL compared with other methods, suggesting that the learned fine-grained prompts in VCF-CLIP significantly improve patch-level visual-text alignment and thereby enhance AL performance.

4) *Complexity and Efficiency Analysis*: To evaluate the practical deployment cost of the proposed method, we compare the complexity and inference efficiency of representative ZSAD methods in Table III. Compared with training-required methods, training-free methods exhibit significantly slower inference, as they typically rely on more complex multi-stage or multiscale inference. All training-required methods adopt CLIP ViT-L/14@336px as the backbone and use an input resolution of 518×518 under their original settings. Among these methods, VCF-CLIP achieves the best overall performance, with 90.6% image-level AUROC and 92.4% pixel-level AUROC, while maintaining a relatively fast inference speed of 124.2 ms per image. In particular, VCF-CLIP is 115.4 ms faster than AdaCLIP and 44.6 ms faster than

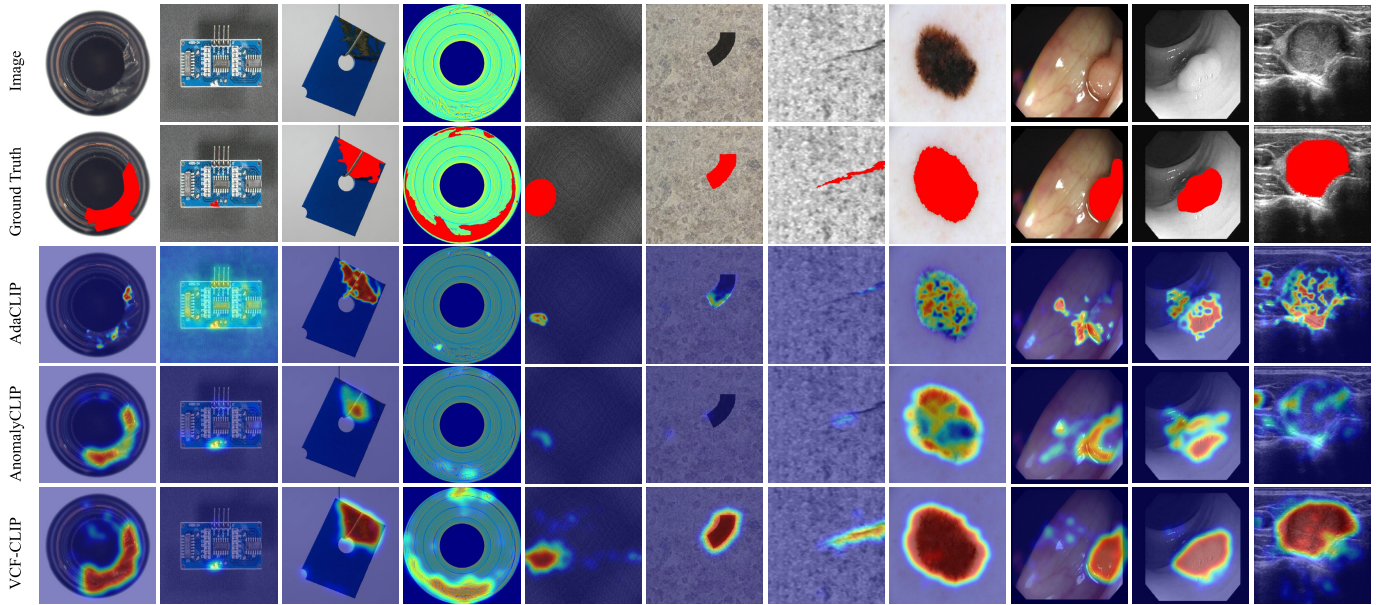


Fig. 4. Visualization of AL results for AdaCLIP [2], AnomalyCLIP [3], and our VCF-CLIP, with samples from left to right drawn from MVtec-AD, VisA, MPDD, BTAD, KSDD, DAGM, DTD-Synthetic, ISIC, ColonDB, ClinicDB, and TN3K datasets.

TABLE III

COMPLEXITY AND EFFICIENCY COMPARISON OF ZSAD METHODS. I-AUROC AND P-AUROC DENOTE THE OVERALL AVERAGE IMAGE-LEVEL AUROC AND PIXEL-LEVEL AUROC, RESPECTIVELY. METHODS MARKED WITH [†] ARE TRAINING-FREE, WHILE THE OTHERS REQUIRE TRAINING

Methods	Backbone	Input Resolution	I-AUROC (%)	P-AUROC (%)	# Params (M)	Inference Time (ms)
SAA [†] [23]	Grounding DINO-SwinT-OGC + SAM-ViT-H	400×400	59.7	73.3	815.9	447.1
WinCLIP [†] [1]	ViT-B-16-plus-240	240×240	82.0	75.4	208.4	274.6
APRIL-GAN [20]	ViT-L/14@336px	518×518	84.3	83.3	<u>432.0</u>	110.6
AdaCLIP [2]	ViT-L/14@336px	518×518	87.8	90.4	439.5	239.6
AnomalyCLIP [3]	ViT-L/14@336px	518×518	89.3	<u>90.7</u>	433.5	168.8
AA-CLIP [58]	ViT-L/14@336px	518×518	90.0	90.4	441.4	128.4
VCF-CLIP	ViT-L/14@336px	518×518	90.6	92.4	444.0	<u>124.2</u>

TABLE IV

ABLATION STUDY ON THE EFFECTS OF DIFFERENT COMPONENTS OF VCF-CLIP USING THE OVERALL AVERAGE AUROC

PPL strategy	PR-Adapter	Pixel-level	Image-level
✓	✓	92.4 (+4.5)	90.6 (+4.5)
✗	✓	91.5 (+3.6)	89.3 (+3.2)
✓	✗	90.8 (+2.9)	88.6 (+2.5)
✗	✗	87.9	86.1

AnomalyCLIP. Overall, these results show that VCF-CLIP achieves a favorable tradeoff between AD performance and practical efficiency.

C. Ablation Studies

1) *Impact of Different Components*: Table IV summarizes the effects of the PPL strategy and PR-Adapter. Without the PPL strategy, we use manually crafted prompts from the AdaCLIP [2]. Introducing the PPL strategy significantly boosts AL and AD performance, increasing pixel- and image-level AUROC by 2.9% and 2.5%. The PR-Adapter, which aggre-

gates visual information into textual embeddings, improves visual-text alignment, resulting in gains of 3.6% and 3.2% in pixel- and image-level AUROC, respectively. The combination of both modules achieves 92.4% pixel-level AUROC and 90.6% image-level AUROC.

2) *Ablation Analysis of the PPL Strategy*: We conduct an ablation study on the PPL strategy, analyzing the effects of prompt length and the parameter λ in the loss function. As shown in Fig. 5(a), both AD and AL performance improve with increasing prompt length but slightly decline when the prompts become too long. While longer prompts capture richer fine-grained information, they may also risk overfitting to the auxiliary dataset, which is consistent with previous findings [21], [22]. Based on the empirical results, we set the prompt length to 16 by default.

Fig. 5(b) shows the impact of λ , which controls the contribution of $\mathcal{L}_{reg}$. Introducing $\mathcal{L}_{reg}$, as opposed to using only $\mathcal{L}_{ali}$, significantly improves both AD and AL performance. This improvement is attributed to $\mathcal{L}_{reg}$ enhancing the distinction between normal and anomalous concepts, resulting in more separable embeddings of normal and anomalous prompts in the latent space. In our experiments, we set λ to 0.1 by default,

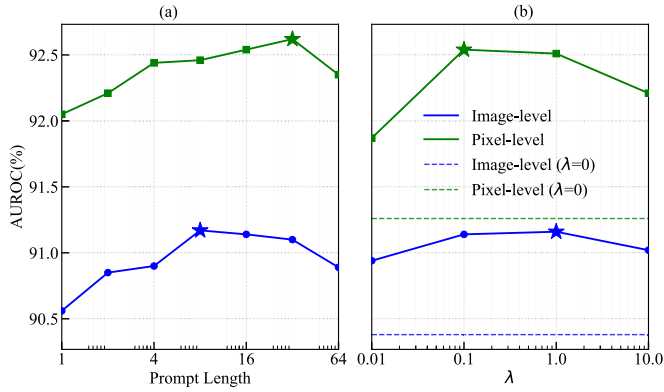


Fig. 5. Ablation of the PPL strategy on the MVTec-AD dataset. (a) Effect of prompt length and (b) effect of λ in the loss function. The best results are indicated by stars.

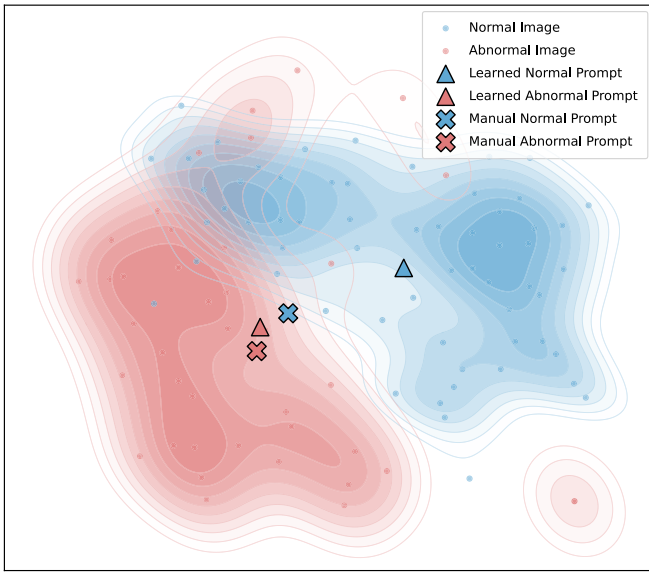


Fig. 6. t-SNE visualization of textual embeddings from learned prompt prototypes and manually designed prompts. The experiment is conducted on the transistor category of the MVTec-AD dataset, where normal and anomalous samples are relatively balanced.

as excessively large values may compromise the pretrained representations of CLIP.

To better understand the learned prompt prototypes, we further visualize their textual embeddings using t-SNE, as shown in Fig. 6. Compared with manually designed prompts, the learned prompt prototypes exhibit better alignment with image embeddings in the latent space. Moreover, the embeddings of manually designed normal and anomalous prompts lie very close to each other, suggesting that CLIP has limited ability to distinguish between normal and anomalous semantics expressed in the natural language, as also observed in previous work [3], [58]. By contrast, the learned normal and anomalous prompt prototypes are separated by a larger margin in the latent space, indicating that they provide a better representation of normal and anomalous concepts and are therefore more suitable for CLIP’s text encoder.

3) *Ablation Analysis of the PR-Adapter*: Table V reports the results of inserting the PR-adapter after different text encoder blocks. Adding a single adapter after the final block

TABLE V
ABLATION STUDY ON PR-ADAPTER PLACEMENT AFTER DIFFERENT TEXT ENCODER BLOCKS USING THE OVERALL AVERAGE AUROC

Blocks	Pixel-level	Image-level	# Adapter Params (M)
—	90.8	88.6	—
4	91.5	89.6	3.02
4, 3	91.9	90.0	6.04
4, 3, 2	92.3	90.5	9.06
4, 3, 2, 1	92.4	90.6	12.08

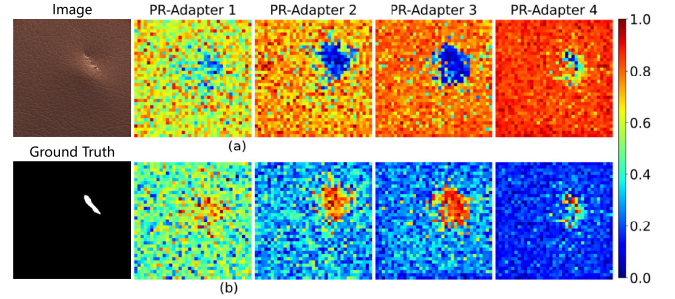


Fig. 7. Visualization of attention maps from different PR-Adapters. (a) Attention maps from the normal prompt. (b) Attention maps from the abnormal prompt.

already yields significant performance improvements at both the pixel and image levels, and progressively inserting adapters into earlier blocks brings further gains. With adapters placed after all four blocks, pixel-level and image-level AUROC increase by 1.6% and 2.0%, respectively, underscoring the benefit of the deeper integration for strengthening visual-text alignment and improving AD and localization. The PR-Adapter is lightweight, containing 3.02-M parameters per instance; inserting four adapters adds 12.08-M trainable parameters, which is only 2.8% of the CLIP ViT-L/14@336px backbone (428.76-M parameters).

We further analyze adapter behavior by visualizing the attention maps in Fig. 7. To capture the overall attention distribution induced by the normal and anomalous prompts, we average the attention weights across all query tokens and heads to obtain a unified spatial map over key locations. In the first PR-Adapter, attention is relatively diffuse, suggesting that it injects basic image information into both the normal and anomalous prompts. As the prompts traverse subsequent adapters, the attention becomes increasingly concentrated, and later adapters progressively route information from normal regions to the normal prompt and from anomalous regions to the anomalous prompt. These observations indicate the effectiveness of the PR-Adapter in refining prompt prototypes through the aggregation of the visual context.

V. CONCLUSION

In this article, we present VCF-CLIP, a fine-grained prompt learning framework for ZSAD that eliminates manual prompt engineering and leverages visual context to refine textual prompts. With the proposed PPL strategy and PR-Adapter, our approach can automatically learn a pair of prompt prototypes and iteratively refine them by aggregating multiscale visual features, thereby generating instance-specific prompts enriched with fine-grained information and achieving superior

visual-text alignment. Extensive experiments across 14 industrial and medical benchmarks demonstrate that VCF-CLIP outperforms state-of-the-art ZSAD methods in both AD and AL tasks. Our method offers new insights into prompt learning and adapter tuning, and we hope it will inspire future research in zero-shot learning and broader vision-language tasks. The limitation of our work is that the learned soft prompts may drift from natural language, which weakens interpretability. In future work, we will extend our framework to additional downstream tasks such as image classification and semantic segmentation to further assess its generalization.

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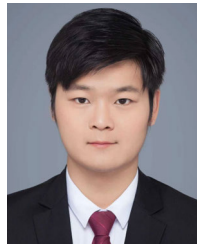
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